

Microbiome

Jan Wälchli

2026-06-08

Table of contents

Background	1
Description all data	2
Sample Removal	2
Summary	2
Sequencing depth plot	3
Rarefaction plot	4
Normalization	4
Kruskal-Wallis test	4
Outlier detection	5
Remaining sample size	5
Phytoplankton	6
Microbial abundances plots	6
Beta diversity	7
PERMANOVA	7
R session information	8

Background

Add your text here

Description all data

Sample Removal

```
[1] "12 samples removed due to low number of sequences (threshold = 390000 sequences)"
```

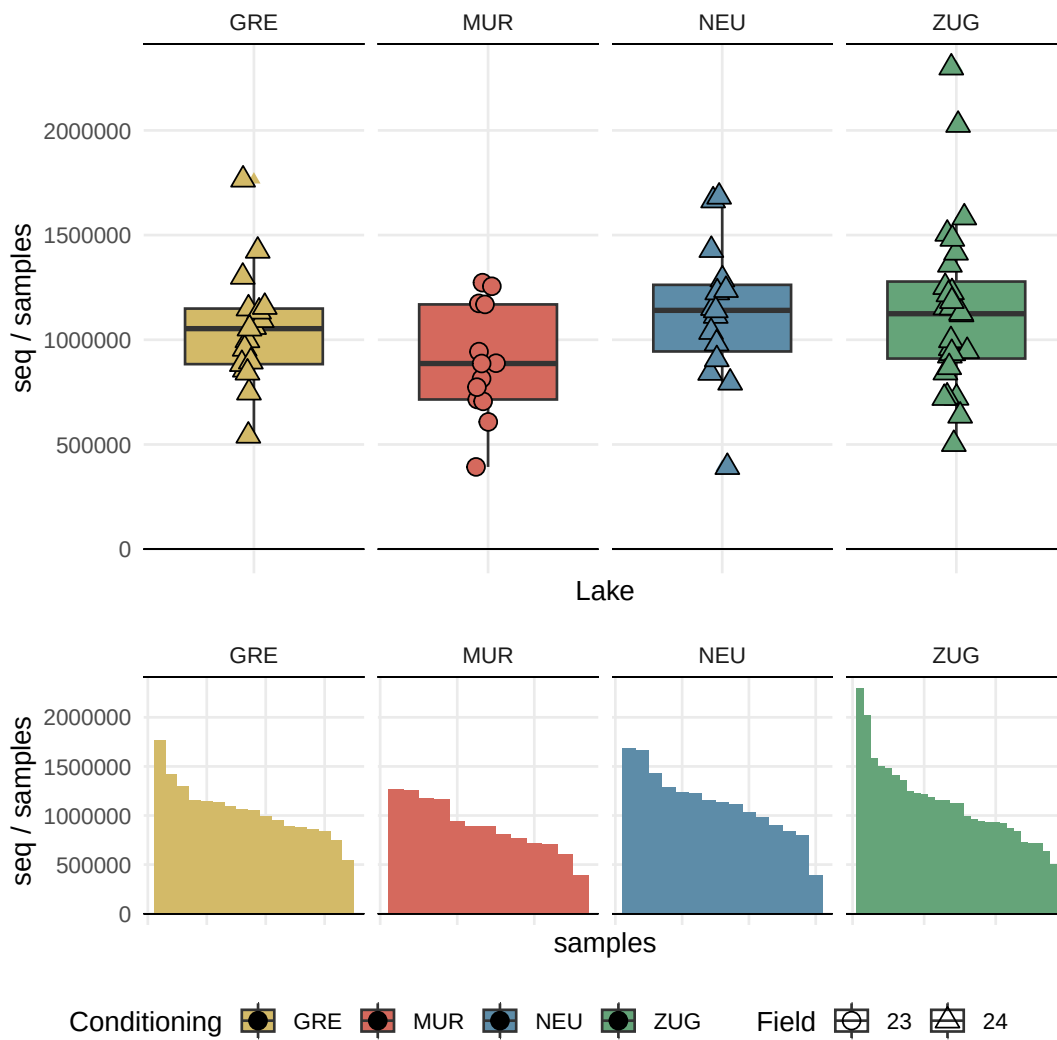
```
[1] "GRE24extractBlank250512"      "GRE24extractBlank250513"  
[3] "MUR23DNAD1Blank2"            "MUR23DNAextractBlank250505"  
[5] "MUR23DNAextractBlank250507"  "NEU24C-DNA-Blank"  
[7] "NEU24extractBlank250514"      "NEU24extractBlank250515"  
[9] "ZUG24N02extractBlank250519"  "ZUG24N02extractBlank250520"  
[11] "ZUG24S03-DNA-Blank"          "ZUG24S03extractBlank250522"
```

Summary

We show the sum, range and median of sequences over all samples.

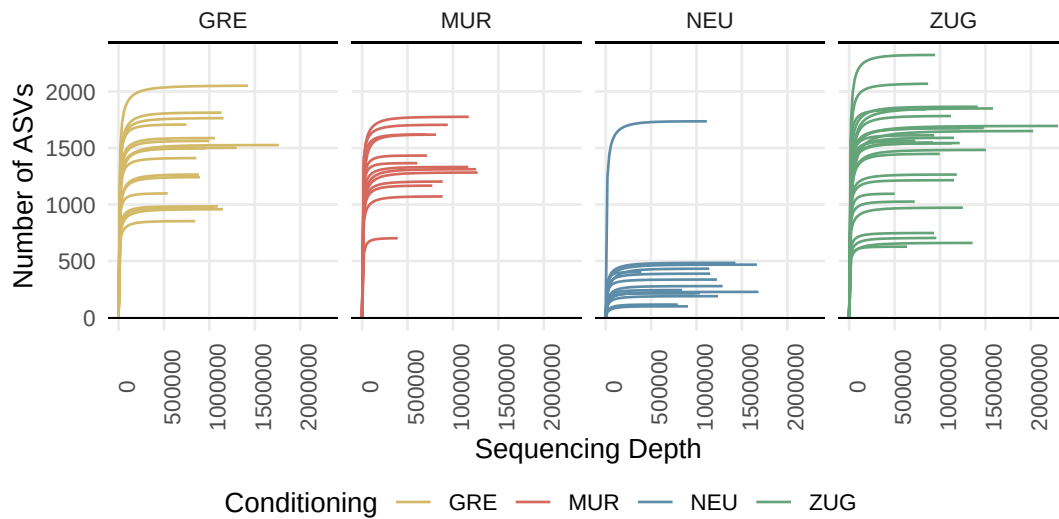
sum	min	max	median	ASVs
78'173'491	391'885	2'299'725	1'053'243	25'912

Sequencing depth plot



Rarefaction plot

We plot a rarefaction plot to check if the sequence depth is enough to capture the microbial diversity.



Normalization

Kruskal-Wallis test

To decide on how to normalize the data we follow the recommendation of Weiss et al. (2017, Microbiome Journal) and inspect whether there are differences in sequencing depths between the different groups(soil-plant-conditioning) by using the non-parametric Kruskal-Wallis Test.

```
Kruskal-Wallis rank sum test
```

```
data: sample_depth by group
```

```
Kruskal-Wallis chi-squared = 4.7194, df = 3, p-value = 0.1935
```

Conclusion: Since we find no significant difference between sample groups we normalise by total sum scaling

Outlier detection

We use the method CLOUD developed by Montassier et al. 2018, which is a non-parametric detection test for outliers. We perform the test with Bray-Curtis distances from the normalized data for each substrate and each plastic treatment individually. We set the number of nearest neighbors to 40% of the samples size and chose an empirical outlier percentile of 0.1. We remove all outliers from our data.

Species	Outliers
Bacteria	1

Remaining sample size

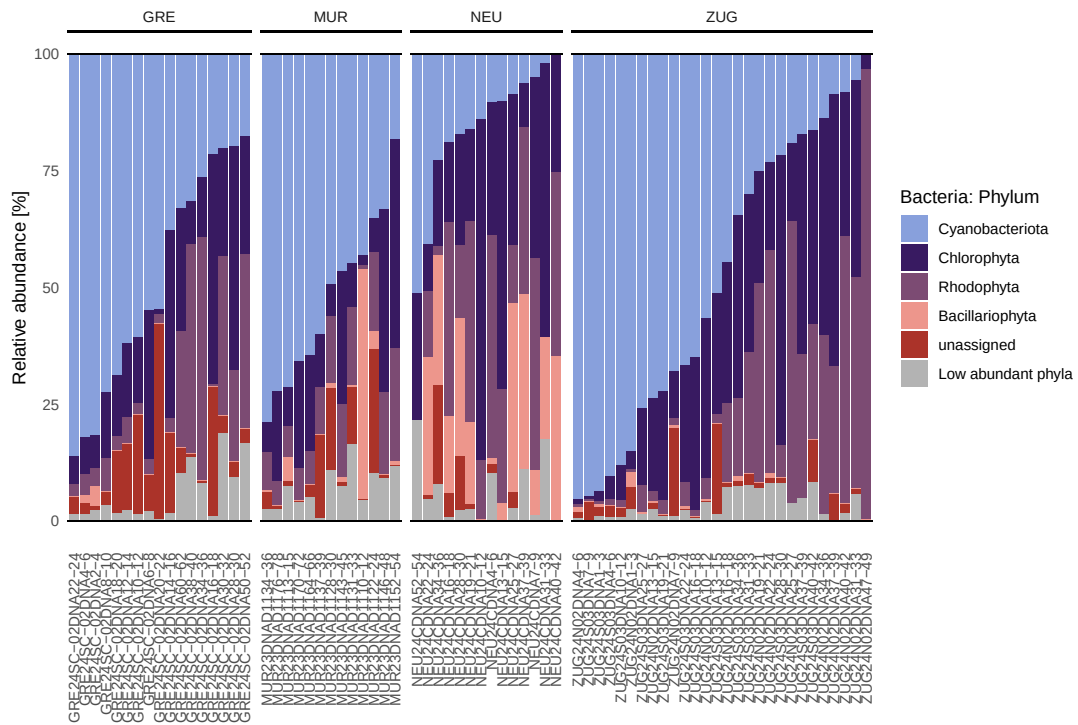
We end up with the following number of samples per treatment for the analysis.

GRE	MUR	NEU	ZUG
17	13	14	28

Phytoplankton

We filter and continue only with phytoplankton classes.

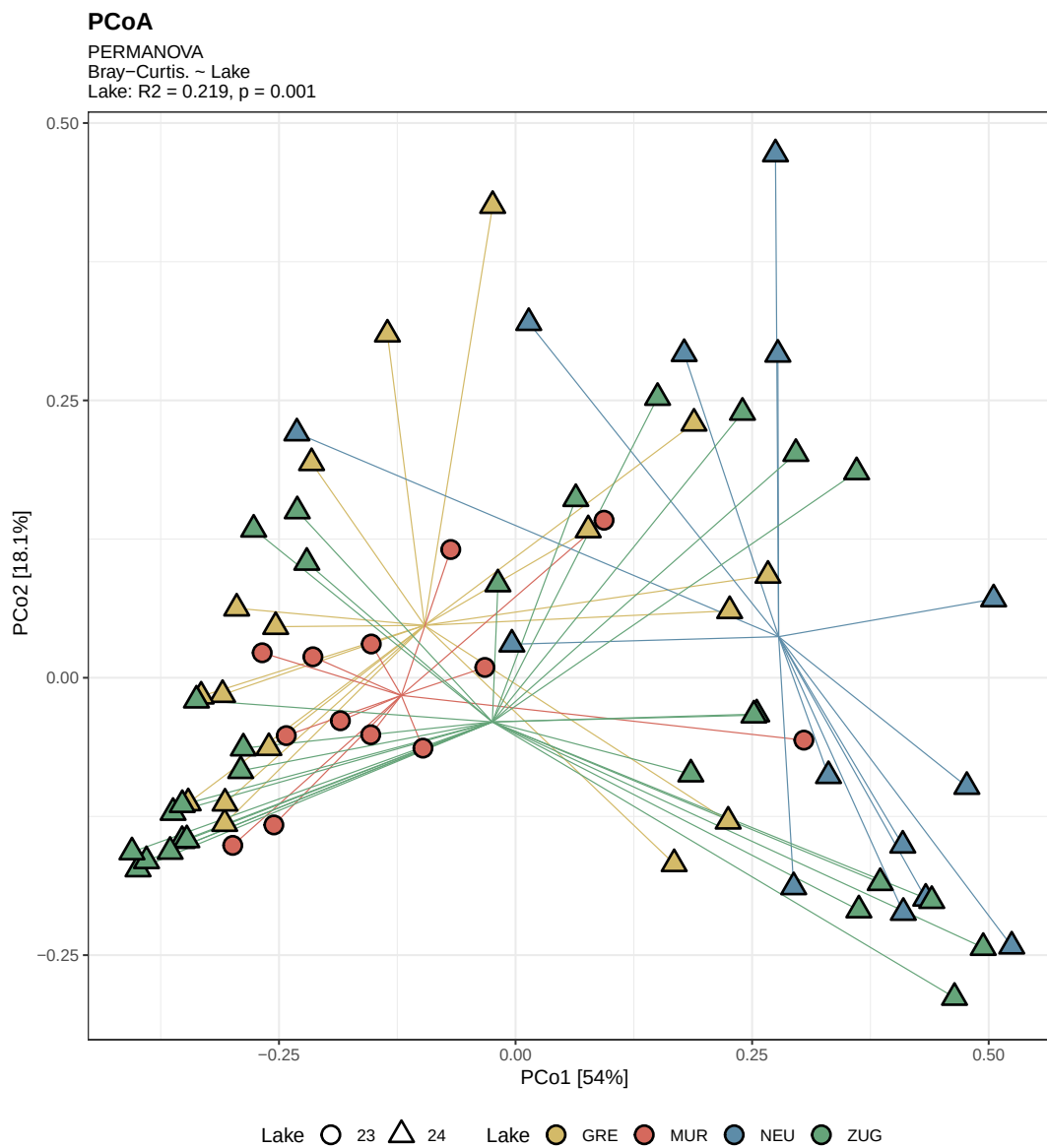
Microbial abundances plots



Beta diversity

PERMANOVA

	Df	SumOfSqs	R2	F	Pr(>F)
Lake	3	6	0.219	6.374	0.001
Residual	68	20	0.781	-	-
Total	71	26	1.000	-	-



R session information

R version 4.5.3 (2026-03-11)
Platform: aarch64-apple-darwin20
Running under: macOS Tahoe 26.5

Matrix products: default

BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

locale:

[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8

time zone: Europe/Zurich

tzcode source: internal

attached base packages:

[1] stats graphics grDevices utils datasets methods base

other attached packages:

[1] dada2_1.38.0	Rcpp_1.1.0	gt_1.1.0	RColorBrewer_1.1-3
[5] ggh4x_0.3.1	cowplot_1.2.0	readxl_1.4.5	vegan_2.7-2
[9] permute_0.9-8	phyloseq_1.54.2	lubridate_1.9.4	forcats_1.0.1
[13] stringr_1.6.0	dplyr_1.2.1	purrr_1.2.0	readr_2.1.6
[17] tidyr_1.3.1	tibble_3.3.0	ggplot2_4.0.3	tidyverse_2.0.0
[21] this.path_2.8.0			

loaded via a namespace (and not attached):

[1] bitops_1.0-9	deldir_2.0-4
[3] rlang_1.2.0	magrittr_2.0.4
[5] ade4_1.7-24	matrixStats_1.5.0
[7] compiler_4.5.3	mgcv_1.9-4
[9] png_0.1-9	vctr_0.7.3
[11] reshape2_1.4.5	pwalgn_1.6.0
[13] pkgconfig_2.0.3	crayon_1.5.3
[15] fastmap_1.2.0	XVector_0.50.0
[17] labeling_0.4.3	Rsamtools_2.26.0
[19] rmarkdown_2.30	tzdb_0.5.0
[21] tinytex_0.58	xfun_0.54
[23] cigarillo_1.0.0	jsonlite_2.0.0
[25] biomformat_1.38.3	DelayedArray_0.36.1
[27] BiocParallel_1.44.0	jpeg_0.1-11

[29] parallel_4.5.3	cluster_2.1.8.2
[31] R6_2.6.1	stringi_1.8.7
[33] GenomicRanges_1.62.1	cellranger_1.1.0
[35] Seqinfo_1.0.0	SummarizedExperiment_1.40.0
[37] iterators_1.0.14	knitr_1.50
[39] IRanges_2.44.0	Matrix_1.7-4
[41] splines_4.5.3	igraph_2.3.0
[43] timechange_0.3.0	tidyselect_1.2.1
[45] rstudioapi_0.17.1	abind_1.4-8
[47] yaml_2.3.10	codetools_0.2-20
[49] hwriter_1.3.2.1	lattice_0.22-9
[51] plyr_1.8.9	Biobase_2.70.0
[53] withr_3.0.2	ShortRead_1.68.0
[55] S7_0.2.1	evaluate_1.0.5
[57] survival_3.8-6	RcppParallel_5.1.11-2
[59] xml2_1.5.0	Biostrings_2.78.0
[61] pillar_1.11.1	MatrixGenerics_1.22.0
[63] foreach_1.5.2	stats4_4.5.3
[65] generics_0.1.4	S4Vectors_0.48.1
[67] hms_1.1.4	scales_1.4.0
[69] glue_1.8.0	tools_4.5.3
[71] interp_1.1-6	data.table_1.17.8
[73] GenomicAlignments_1.46.0	fs_1.6.6
[75] grid_4.5.3	ape_5.8-1
[77] latticeExtra_0.6-31	nlme_3.1-168
[79] cli_3.6.5	S4Arrays_1.10.1
[81] gtable_0.3.6	digest_0.6.38
[83] BiocGenerics_0.56.0	SparseArray_1.10.10
[85] farver_2.1.2	htmltools_0.5.8.1
[87] multtest_2.66.0	lifecycle_1.0.5
[89] MASS_7.3-65	